# Management of Normal Labor

## Objectives

**At the End of this Lecture Each Student should be able to: -**

1-Discuss factors affecting the decision of admission.

2-Obtain complete history of the mother before admission.

3- Performe examination of the abdomen.

4-Performe examination of the vagina

5- Obtain laboratory tests on admission.

6-Apply management of the first stage.

7-Prepare the delivery room.

8-Understand duties of the scrub nurse.

9- Understand duties of the circulating nurse

10- Apply management of second stage of labor.

11- Apply management of third stage of labor

12- Apply management of fourth stage of labor

## Admission to labor unit

**Factors Affecting the Decision of Admission**

- Frequency of contractions (3/10 admit)
- Prior vaginal deliveries (admit)
- Cervical dilatation (>4cm admit)
- Condition of membranes (rupture admit)

## Obtaining the Admission History and Performing Examinations

## Identifying Information

- Name
- Age
- Duration of marriage
- Educational level
- Last meal since

## History of Past Pregnancy

- Mode of delivery
- Size of infant
- Birth intervals
- Duration of pregnancy
- Congenital defects
- Any abnormality affecting that pregnancy.

## Medical and Family History

- Major medical and or surgical problems
- Drug allergies

## History of Current Pregnancy: -

- Prenatal care
- Laboratory tests
- Special tests as amniocentesis, NST.
- Details of any complications or problems and their treatment

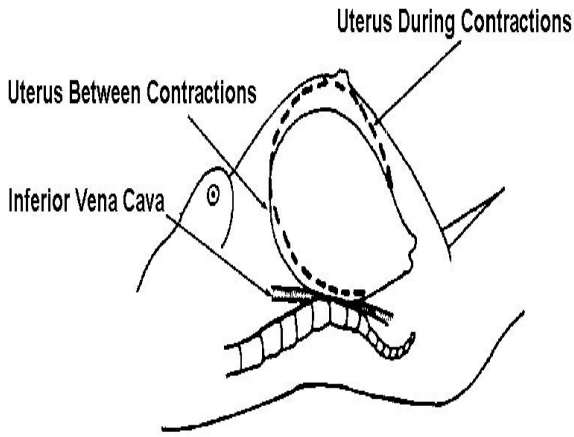
## Obstetric History

- LMP (Last Menstrual Period)
- EDD or EDC (Estimated date of confinement)
- Gestational age
- Parity, gravida.
- Last labor since
- Mode of previous delivery
- Any Medical/ OB complications has attacked the present pregnancy.
- Fetal condition (Biophysical Profile)

## Examination of the Abdomen

- Inspect the contour of the abdomen.

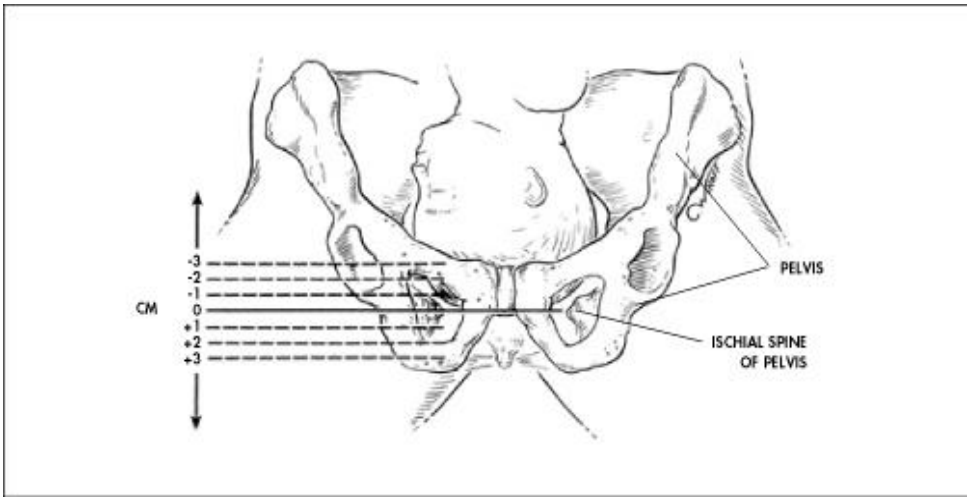
- Palpate the abdomen gently with warm hands, for the height of the fundus, lie, presentation, position, attitude and station then record them.
- Observe and record the frequency, duration, and intensity of uterine contractions. The purpose to assess the ability of the uterus to dilate the cervix, determining the progress of labor, help to detect abnormalities of uterine contractions (such as lack of uterine relaxation), and help to evaluate any signs of fetal distress.
- Auscultate and record the fetal heart sound for one minute using Pinard's or Sonicade.




**Palpation of uterine contraction**

## Examination of the Vagina

1. Assess the progress of labor.
2. Determine the extent of effacement and the degree of dilatation of the cervix.
3. Assess the condition of the membranes.
4. Determine position, presentation, and degree of descent of head in pelvis in relation to ischial spine (station).
5. Observe the presence of umbilical cord beside the head.
6. Check moulding, presence of caput succedaneum.
7. Evaluate pelvic capacity




## Obtaining Laboratory Testes

1- ABO/ Rh.

2- CBC

3- Urine analysis: measure for glucose and proteinuria.

## General Management of the first stage of labor

### 1. Asepsis:

The danger of infection has been minimized using antibiotics; it is very important to take all antiseptic measures.

### 2. Examination of the patient:

- General, abdominal, and vaginal examinations are carried out.
- Urine sample is taken for sugar and albumin.

### 3. The urinary bladder:

Kept empty by frequent micturition or passing a catheter at strict antiseptic measures.

### 4. The rectum:

Evacuated by low enema (warm water), to avoid inertia and soiling of the perineum by feces during delivery (new trends) ----- must be avoided.

**Some physicians consider giving enema to:**

a) Prevent fecal contamination of the perineum during delivery.

b) Cleanse the bowel. This provides more room for fetal passage.

c) Stimulate uterine contractions.

**Contraindication of Enema**

(a) Vaginal bleeding.

(b) Premature labor.

(c) Presenting part not engaged.

(d) Abnormal presentation-----breech or transverse.

(e) Advanced labor.

(f) Membranes are ruptured or danger of prolapsed cord.

**5. Patient position:**

A. The patient is allowed to walk if the head is engaged.

B. Rest in bed is recommended when membranes are ruptured, or the head not engaged.

**6-Diet:** Nothing is given by mouth except sugary fluid to avoid vomiting. I.V

Glucose is given in prolonged labor.

**7. Sedative and analgesics:** To avoid exhaustion from pains e.g. pethidine 50 mg IM

**8. Observations:**

**a. Maternal observations**

C. Pulse, temperature, blood pressure every hour

D. Uterine contractions: Frequency and duration from ½ - 1 hour

E. Time of rupture of membranes

F. Degrees of cervical dilatation

**b. Fetal observations:**

G. Fetal heart sound: every 30 minutes

H. Degrees of head descend.

## 9-Nonpharmacological interventions

I. Assist client to breathe through each contraction.

J. Assist with relaxation techniques between contractions to avoid maternal exhaustion.

K. Provide encouragement to client during labor.

L. If possible, have a family member or friend stay with client during labor.

M. Provide sips of water or juice during early labor.

N. If client is unable to tolerate fluids orally, provide mouth care with water swabs to coat the lips.

## Nursing Care during First Stage Labor

1 - Assessment

2 -Physical Care

3 -Emotional/Supportive

### 1-Assessment

- Blood Pressure, Pulse & Respirations – q 30-60 min.
- Temperature – q 4 hr (unless ROM then q 1 hr)
- Uterine activity – q 30-60 min.
- Fetal Heart Rate – q 30-60 min.
- Vaginal show – q 30-60 min.
- Behavior – q 30 min.
- Vaginal examination – as needed to document progress

### 2-Physical Care

-Nutrition: IV glucose or ringer lactates to prevent dehydration

- Nonpharmacological interventions:

- Assist client to breathe through each contraction.
- Assist with relaxation techniques between contractions to avoid maternal exhaustion.
- Provide encouragement to client during labor.
- If possible, have a family member or friend stay with client during labor.
- Provide sips of water or juice during early labor.
- If client is unable to tolerate fluids orally, provide mouth care with water swabs to coat the lips.

**Physical sensations during latent phase**

- Thirst, anorexia, nausea, vomiting, diarrhea
- Uterine contractions usually become more frequent and intense as
- latent labor progresses; felt in the abdomen and lower back.
- Fetal descent may cause pressure on bladder, creating urge to urinate
- Fatigue if latent labor is prolonged.

**3- Emotional Support**

**Latent Phase**

1. Provide current status and process of labor.
2. Demonstrate breathing and relaxation techniques and work with support person.

**Active Phase**

3. Feedback about performance; work with support person
4. Reduce distractions (quiet)
5. Review current status and process of labor.
6. Position changes

## Transition Phase

7. Client may lose control; provide continuous support.
8. Reduce distractions (quiet)
9. Help to relax between contractions.
10. Inform client about short duration of transition phase.
11. Keep informed of progress.

## Management of the second stage of labor

Assessment is continuous during second stage labor

| Assessment Item | Frequency |
|---|---|
| Vital signs (BP, P, R) | q 5-30 minutes |
| Uterine activity | every contraction |
| Bearing down effort | every contraction |
| Fetal heart rate | q 5 minutes |
| Signs of fetal descent | q 10 minutes |
| Behavior, mood, etc | q10-15 minutes |

## Preparation of the delivery room

1- The delivery room should be warm

2- Enough privacy should be provided

3- Delivery equipment (2 kokher, 2 arteries, 1 needle holder, catgut 2 scissors, 1tissue forceps, drugs as methergine and oxytocin,)

4- Resuscitation equipment

5- Warm Incubator

6- Oxygen supply

7- Infant suction apparatus

8- Infant resuscitation equipment as bag and masks.

## Preparation of the woman

1-Transfer to delivery room, should be between contractions and supported adequately (primipara when cervix is fully dilated; multipara when cervix is 8cm).

2- Posture of the woman (dorsal or lithotomy; left lateral)

3- The legs and thighs should be dressed in sterile leggings

4- Sterile towels put over the abdomen and under buttocks leaving only the vulva and perineum exposed.

**Delivery positions**

**Dorsal lithotomy:** provides maximum perineal exposure.

**Preparation of the attendance**

1-The attendance should wear a head cap and mask.

2-Wash and scrub hands

3- Put on a sterile gown and gloves.

**Duties of the scrub nurse**

1-Arranges the instruments on the delivery table

2- Assists in draping the woman

3- Handles the instruments and catheter to the doctor as called for

4- Assists in perineal repair and prepares proper sutures on proper needles

5- Massages the uterus as needed

6- Assists in holding the newborn infant.

**Duties of the circulating nurse**

1-Assists the woman during her transfers from the labor bed and positions her properly on the delivery table

2- Opens packs

3- Prepares and administers routine medications as ordered and records its time

4- Prepares resuscitation and emergency equipment.

5- Count and records FHR

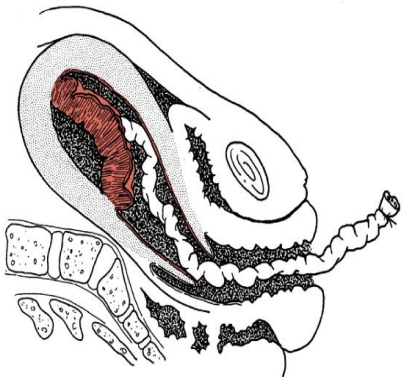
6-Receives the infant from the doctor and takes care of him

7- Records the time of birth and sex of the infant

8- Observes and records the time of placental expulsion

9- Observes the location and extent of episiotomy or tear

10- Records the amount of blood loss



11- Observes and records any unusual conditions

12- Prepares and places the identification measure of the infant

13- Assists in giving anesthesia and emergency drugs if needed

## Care during delivery

1-Encourage mother to take deep breath, holds it and bear down during contractions and to relax between contractions.

2-Support perineal to prevent laceration

3- Once head is delivered, check for presence of cord around neck

4-If cord is wrapped around neck; gently slip a finger under cord and pull

1- Apply two forceps on umbilical cord and cut in between

2- Carefully inspect vagina, perineum and labia for tears or lacerations

3- Inspect vulva for bleeding, edema and hematoma

4- Make sure that the uterus is well contracted.

5- Avoid vigorous uterine massage.

## C. Management of the third stage

### Signs of placental separation:

The hand is placed over the fundus to detect signs of placental separation and descend.

A. The body of uterus becomes smaller, harder and more globular (the earliest sign)

B. The fundus rise up because the LUS now distended by the placenta

C. A supra-pubic bulge due to presence of placenta in LUS

D. Elongation of the cord outside the vulva

E. Sudden gush of blood from the vagina

F. No pulse in umbilical cord

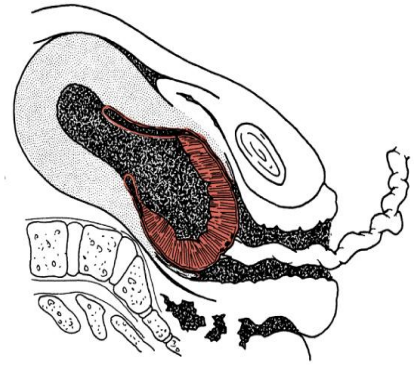
Repair of episiotomy and any perineal lacerations immediately after placental delivery

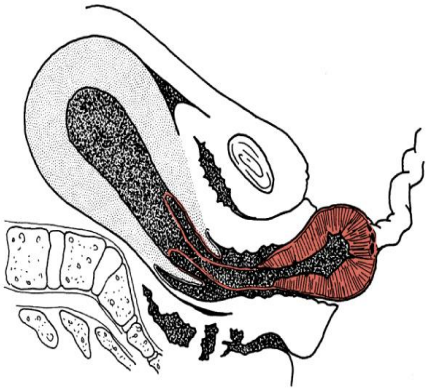
1- The placenta is expelled by controlled cord traction and use of oxytocic drugs

2- Receive placenta in the hands and turns through a circle

3- Examine the placenta and membranes for general appearance, completeness and weight

4- Clean and dry the vulva, buttocks and thighs then apply a sterile pad.





## Inspect the Placenta

- Make sure it is complete.
- Look for missing pieces.
- Look for malformations.
- Look for areas of adherent blood clot.

## Nursing role in immediate newborn care

The main nursing managements are:

- Clear airway.
- Maintain of body temperature.
- Cutting of cord
- Eye care
- Identification.
- Apgar Score

## Immediate need for resuscitation

Implications of Apgar score

★ 0 – 3 Severe distress. Requires emergent resuscitation measures.

★ 4 – 6 Moderate difficulty. Requires supplemental oxygen and appropriate resuscitation measures.

★ 7 – 10 No difficulty. Is considered normal.

## Management of the Fourth Stage of Labor

1-Check and record maternal vital signs every 15 minutes

2-Check the uterus to ensure that it is well contracted to prevent bleeding

3-Observe and record the amount of bleeding

4-Inspect the perineum for edema or hematoma

5-Encourage the woman to pass urine

6-Keep woman dry and clean

7-Light snack and drink

8-Encourage rest and sleep